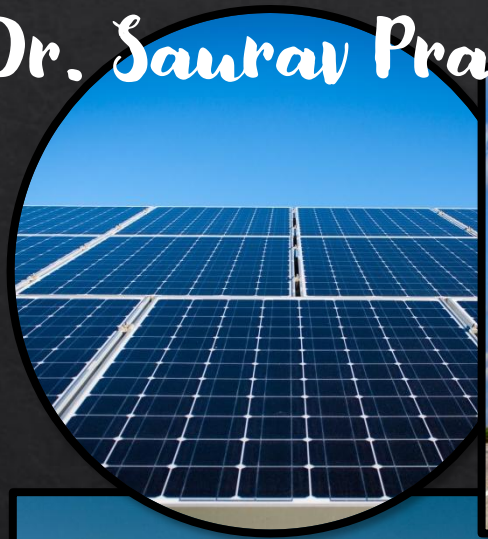
Module 5:
Energy Resources and Sustainability

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VIT BHOPAL
www.vitbhopal.ac.in
A Place to Learn, A Chance to Grow










NATURAL RESOURCES



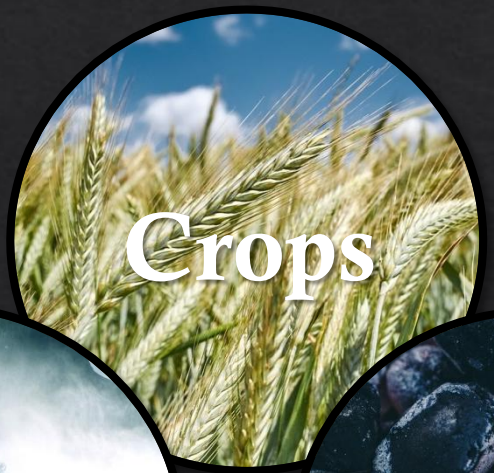
What is a natural resource?

What natural resources do you know?

NATURAL RESOURCES

Natural resources – natural substances required by humans for different needs.













ENERGY RESOURCES

What is an energy resource?

What energy resources do you know?



ENERGY RESOURCES

Natural resources such as **oil, gas, coal, wood, wind, sunlight** and **waves** can be used to **produce energy.** These are **energy resources.**











NON-RENEWABLE & RENEWABLE

What is a non-renewable energy resource?





What is a renewable energy resource?



NON-RENEWABLE & RENEWABLE

Non-renewable energy sources are things like oil, natural gas and coal. They cannot be easily replaced, because they have taken millions of years to form.



We are using them faster than they are being made. This means that one day they will run out!



Renewable energy resources like wind power, wave power, solar power and biofuel can be easily replaced and will not run out.

NON-RENEWABLE ENERGY



Natural gas

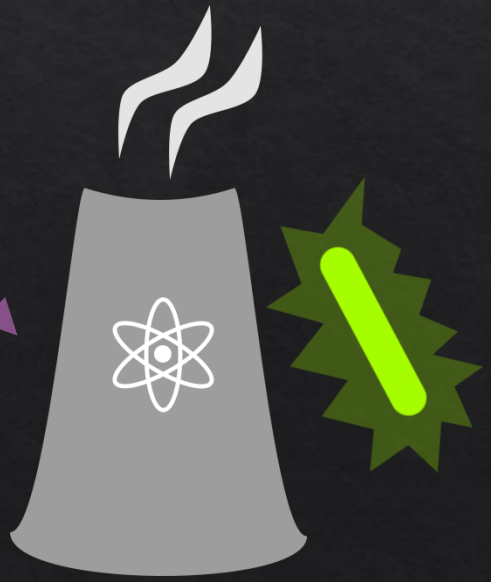
Oil
(petroleum)



Non- renewable



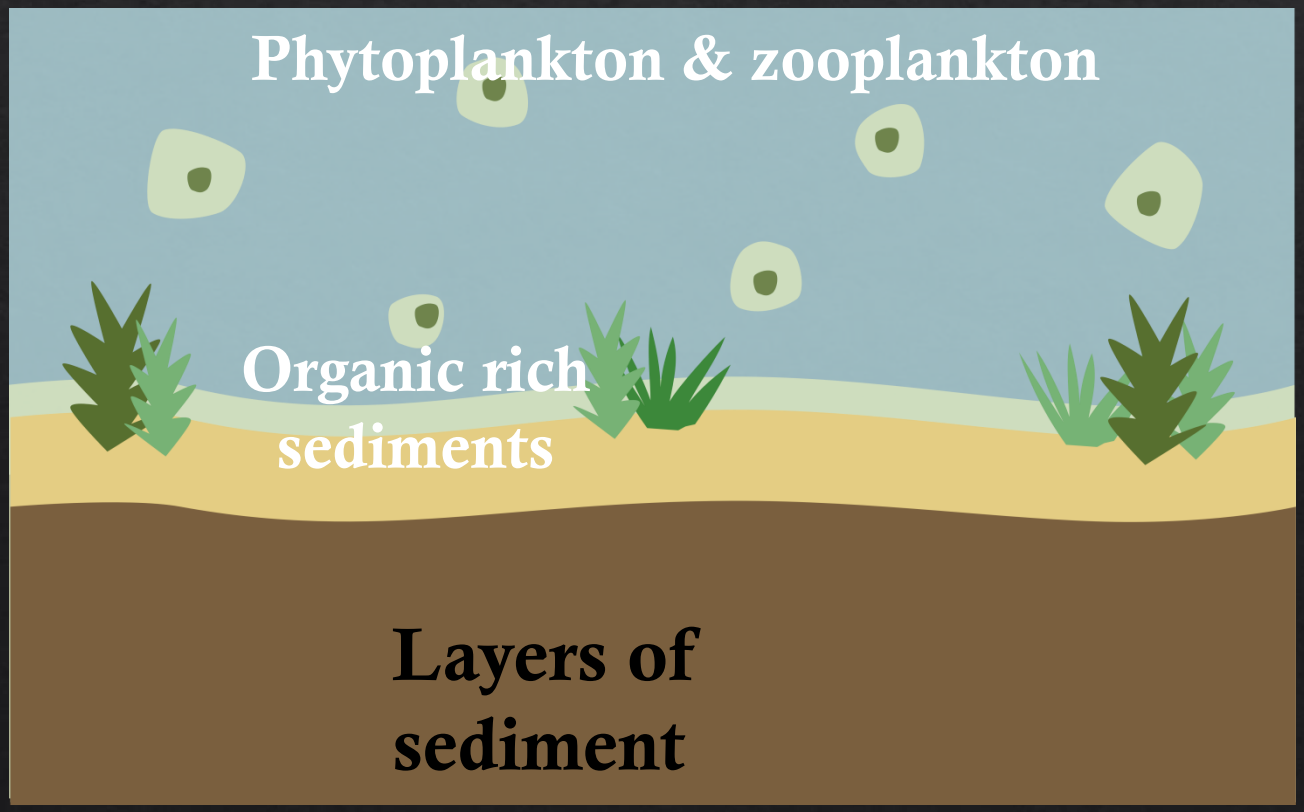
Coal



Nuclear power

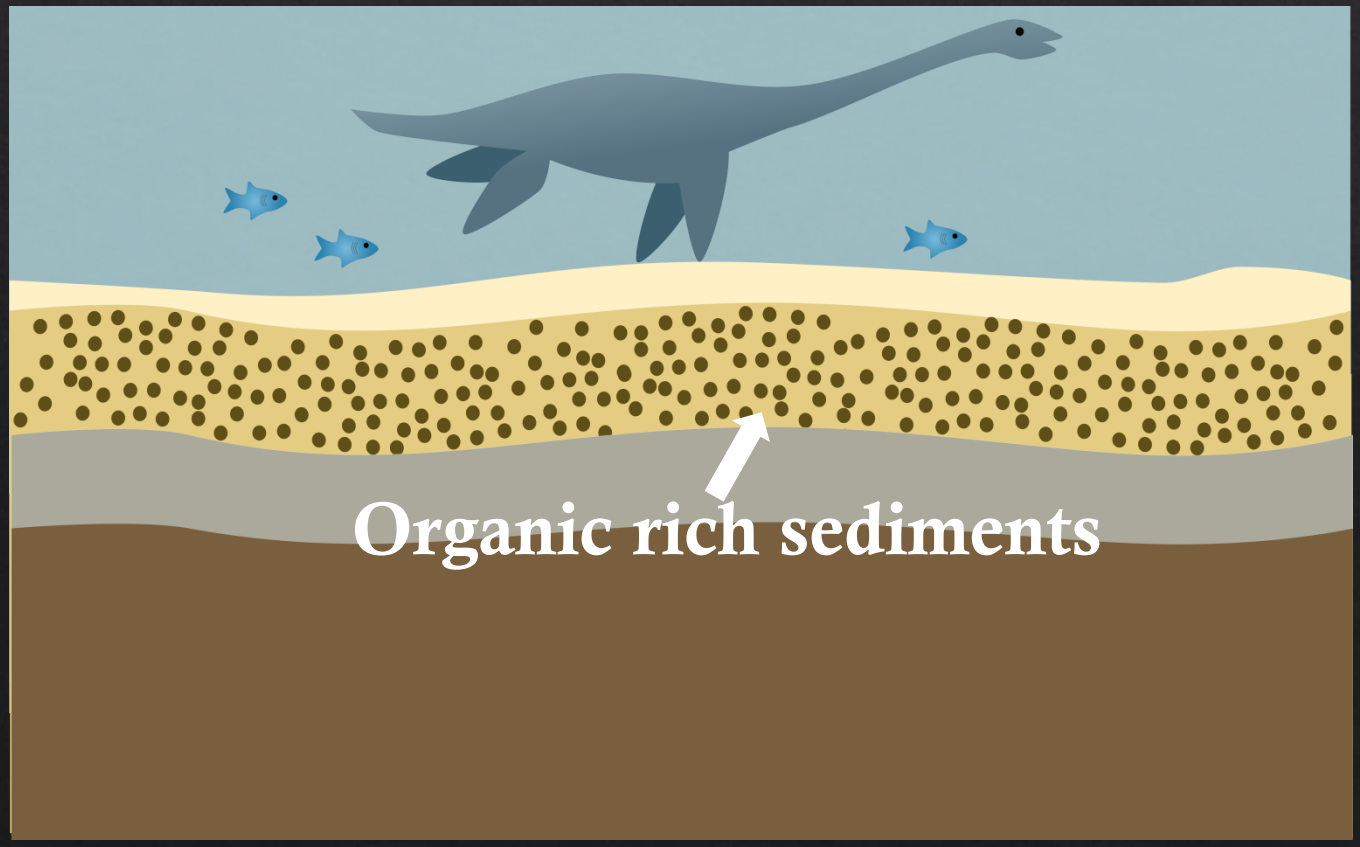
NON-RENEWABLE: OIL & GAS FORMATION

Earth's crude oil and natural gas is formed from marine plants and animals that died millions of years ago



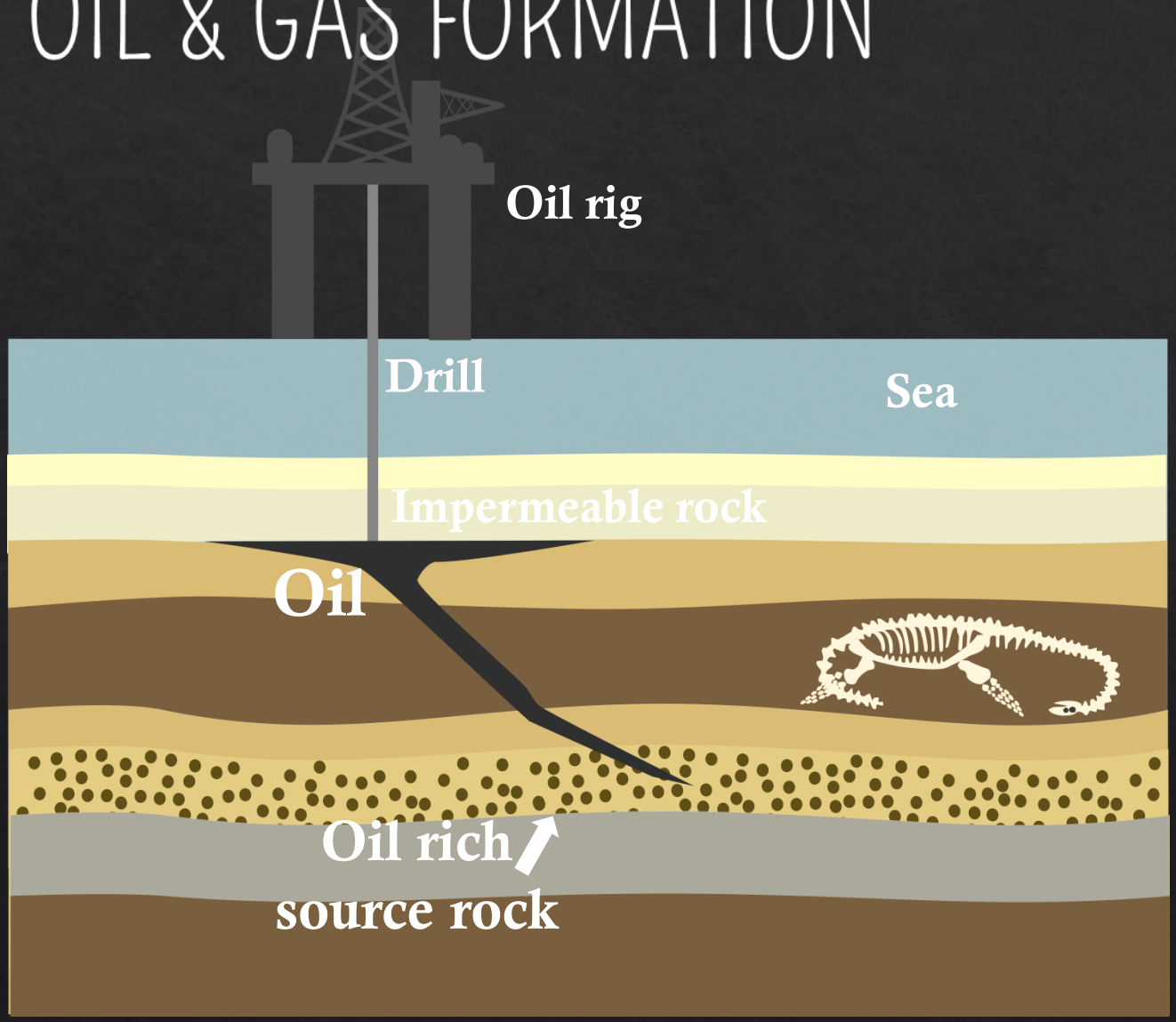
1. Dead microscopic animals and plants sink to the bottom of the ocean and become buried with other sediments (sand and mud)

NON-RENEWABLE:
OIL & GAS FORMATION



2. Over millions of years these sediments get buried deeper and deeper. Temperature and pressure rises and the microorganisms in the sediment are turned into gas and oil

NON-RENEWABLE: OIL & GAS FORMATION



3. Oil and gas move upwards through tiny pores in the rock. Oil and gas gets trapped under impermeable layers of rock and can be then be taken out using long powerful drills.

NON-RENEWABLE: OIL & GAS









Oil & gas → **burned to heat water** → **steam propels turbine** → **Electricity**



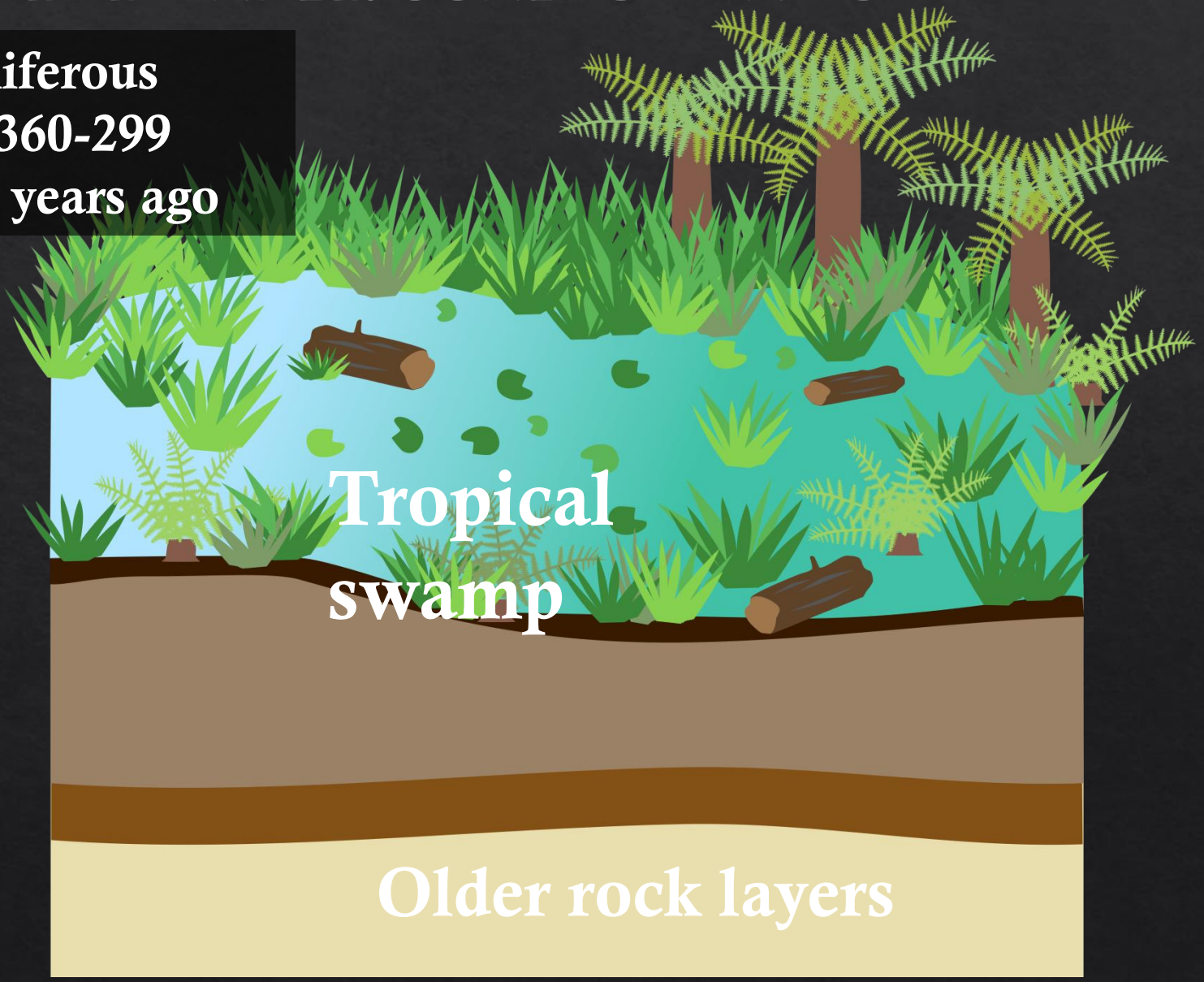
- Crude oil is turned into jet engine oil, petrol and diesel which are used as a **fuel** in planes and cars.



- When oil and gas are burnt they release **carbon dioxide** gas into the atmosphere. CO_2 is a **greenhouse gas** and contributes to global warming.

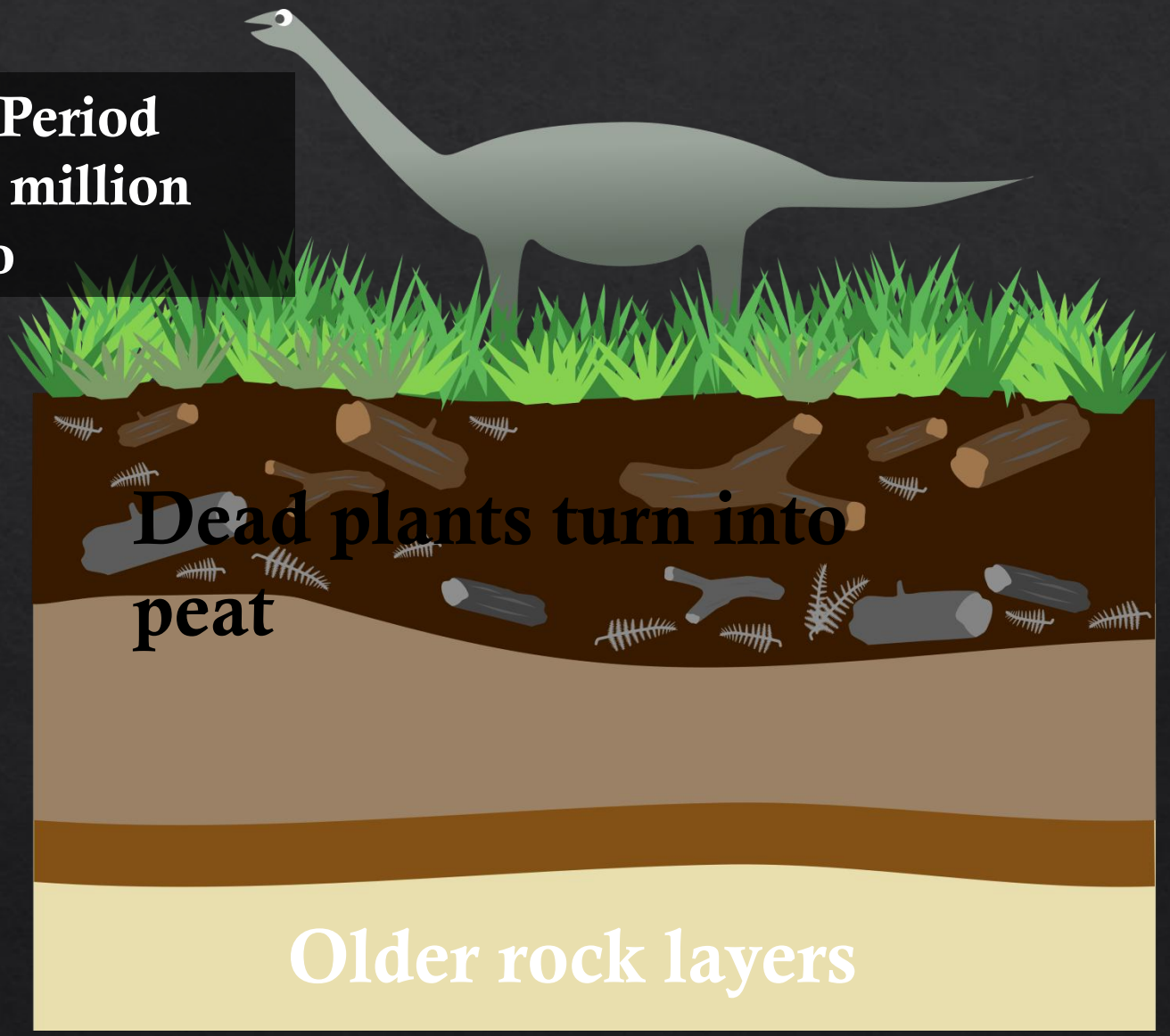
NON-RENEWABLE: COAL FORMATION

Carboniferous Period 360-299 million years ago

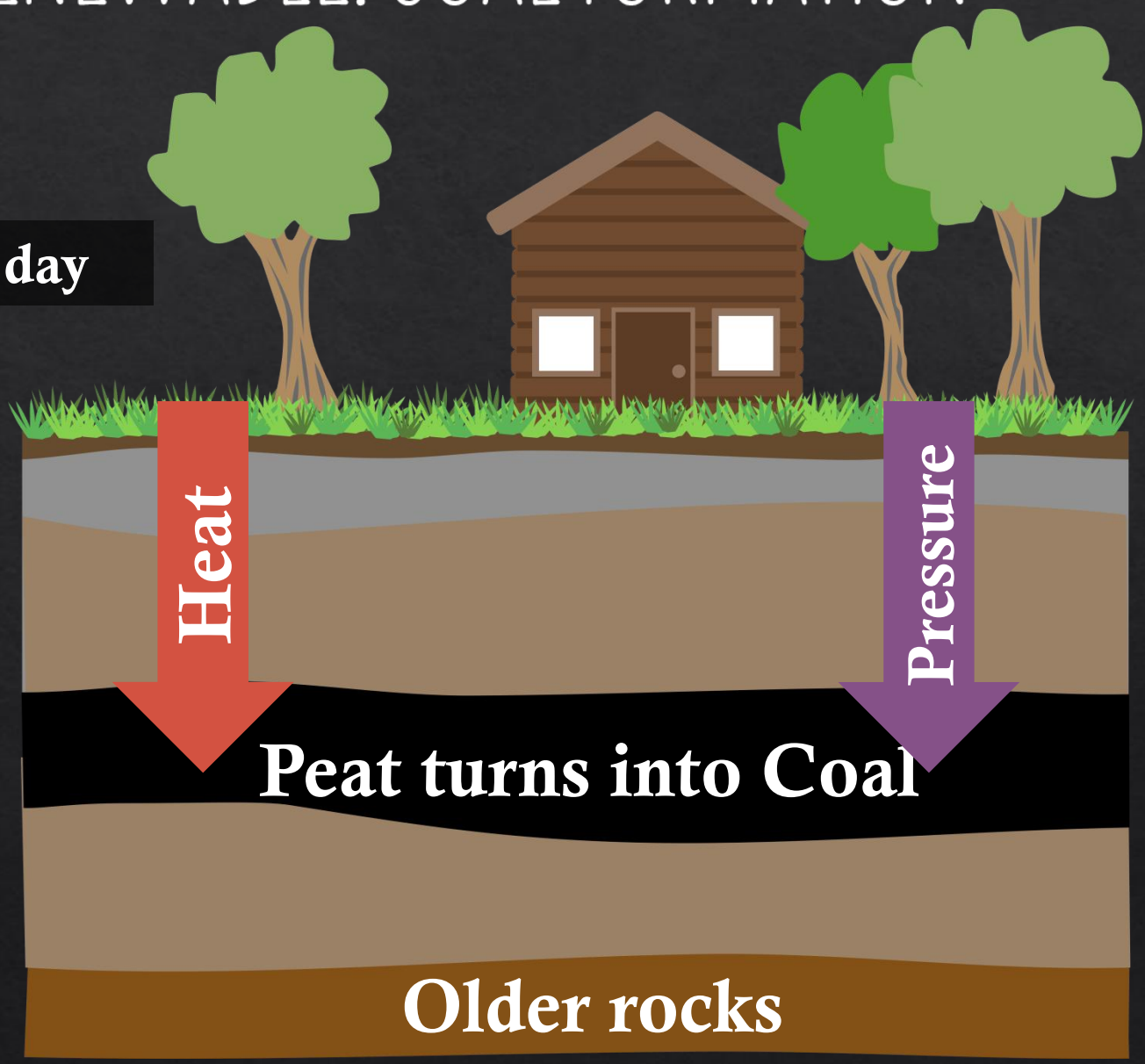


NON-RENEWABLE: COAL FORMATION

Jurassic Period
201- 145 million years ago



NON-RENEWABLE: COAL FORMATION



NON-RENEWABLE: COAL

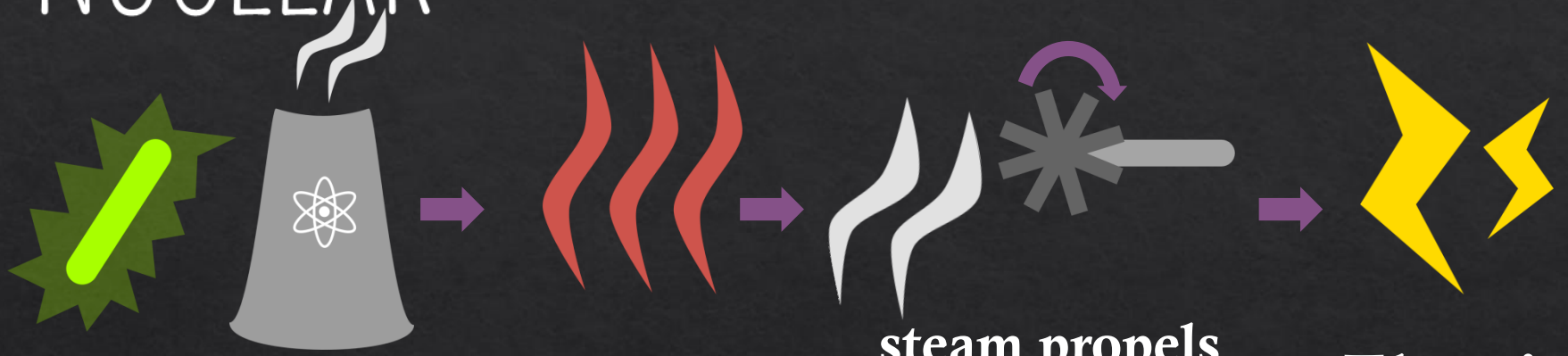


Coal → **burned to heat water** → **steam propels turbine** → **Electricity**

Coal ↓ **burned for heat**

- Coal is **cheap** and there is **lots of it**!
- When coal is are burnt it releases **carbon dioxide, sulphur dioxide** and **nitrogen dioxide** into the atmosphere contributing to global warming and acid rain.

NON-RENEWABLE: NUCELAR



- **Uranium** and **plutonium** are nuclear fuels
- Nuclear fuels do not produce harmful greenhouse gases
- Nuclear power is very **efficient**
- Nuclear power produces **radioactive waste** which is dangerous and has to be seal in containers and buried for thousands of years

RENEWABLE ENERGY



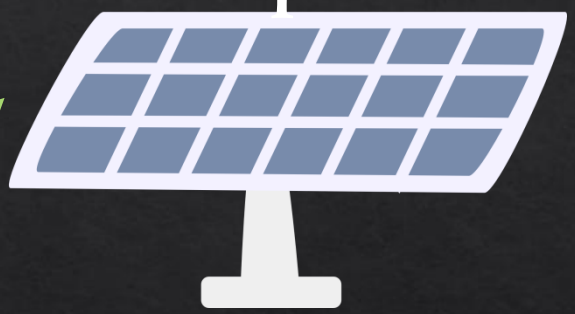
Geothermal



Biofuel



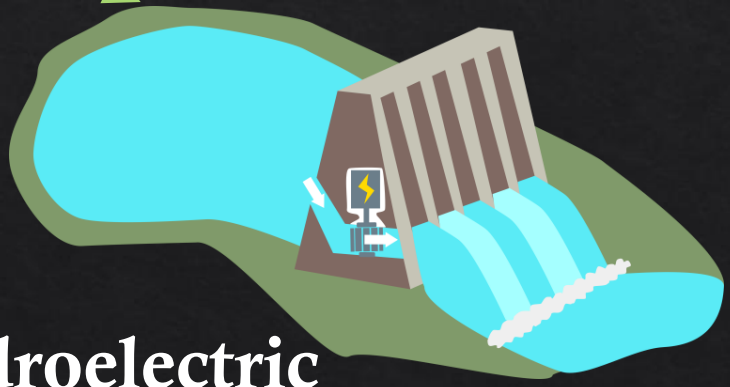
Sola
r



Renewable



Wind



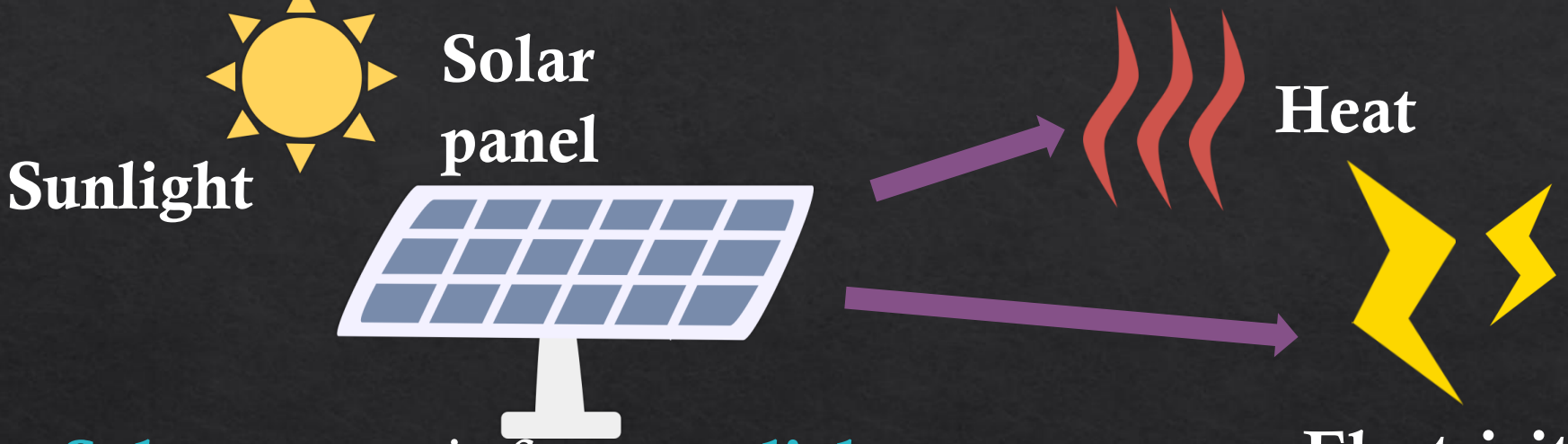
Hydroelectric

RENEWABLE: WIND

- Large windmills are called **wind turbines**
- When the wind blows, the blades move and this spins a **turbine** which is connected to a **generator** which generates **electricity**.
- **Wind farms** are made up of lots of wind turbines this allows lots of electricity to be generated
- Wind farms are built in **windy areas**
- No wind = no electricity



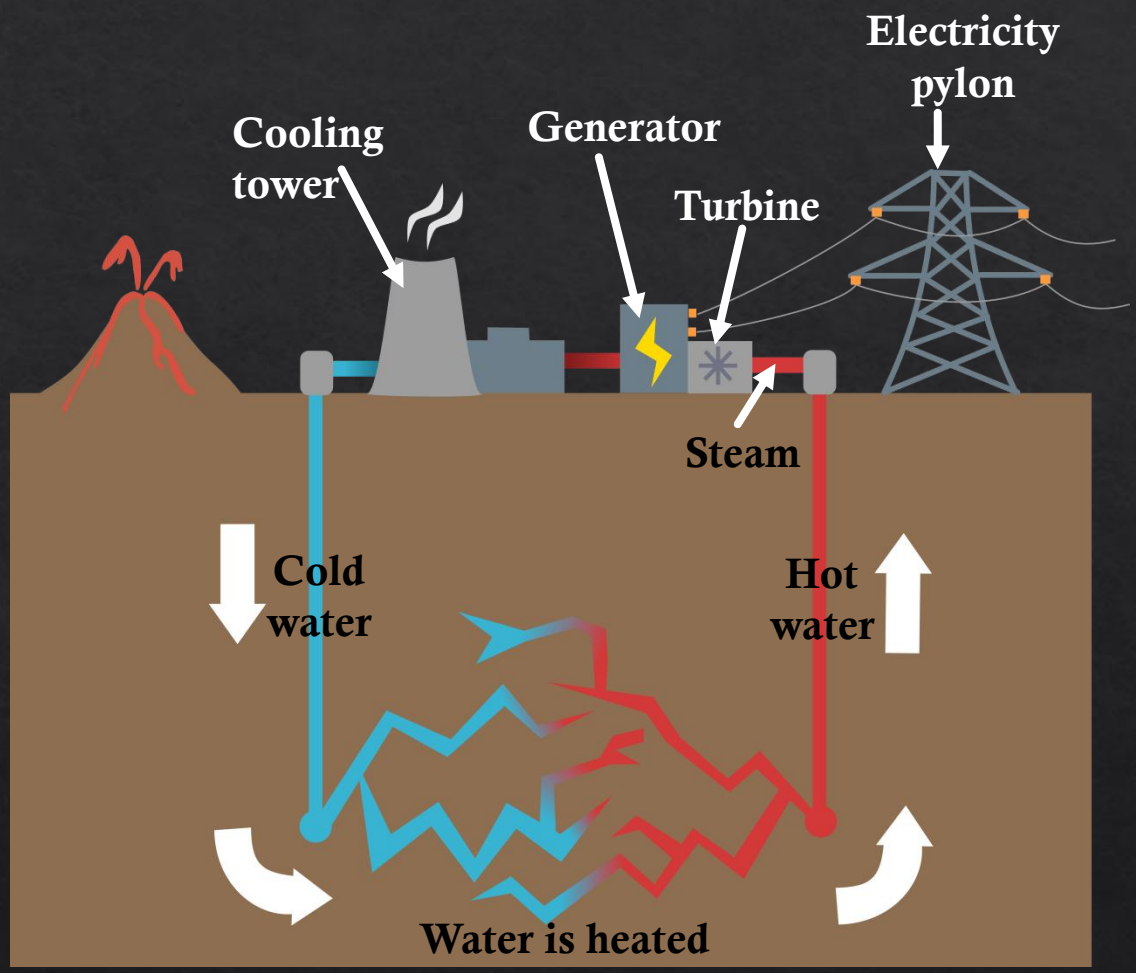
RENEWABLE: SOLAR



- **Solar energy** is from **sunlight**
- Solar energy can be used to generate **heat or electricity**
- **Solar panels** filled with water are used to produce heat
- **Photovoltaic cells** are used to turn sunlight into electricity
- Solar panels can be put of building roofs
- Solar panels are **expensive** and you need a lot of them!

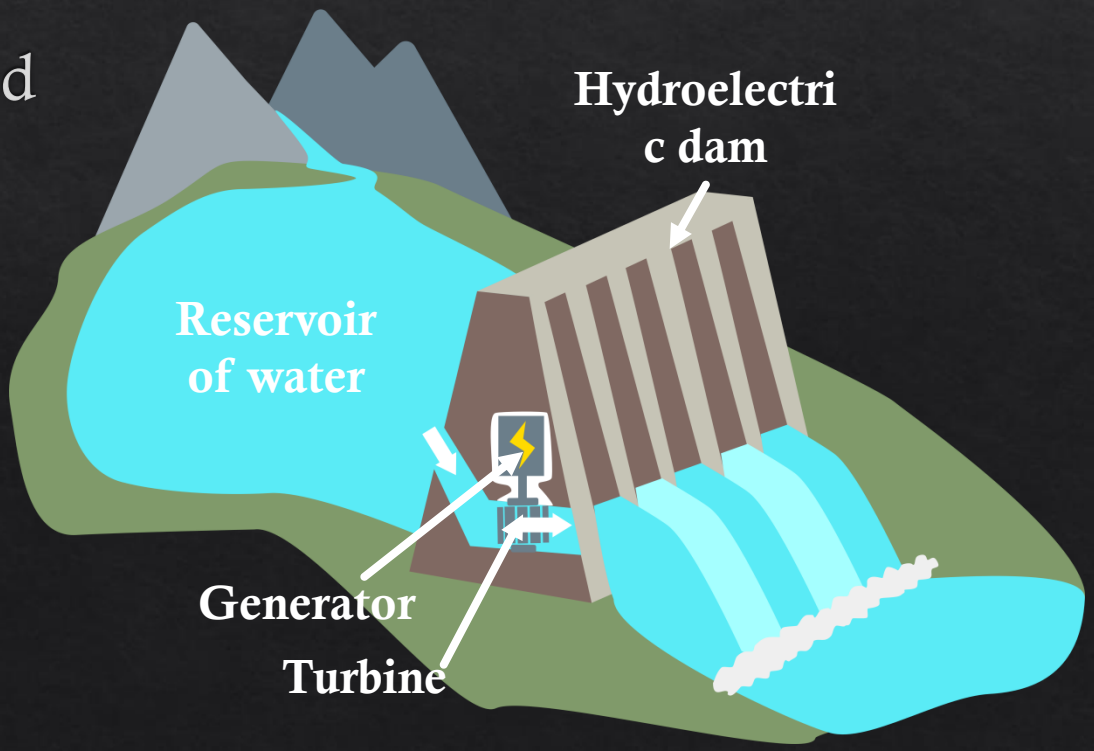
RENEWABLE: GEOTHERMAL

- **Geothermal energy** is heat energy from the Earth
- Water can be **pumped down** into hot rock where it is heated.
- Geothermal energy can be used to produce **heat or electricity**
- **Iceland** is very volcanic so uses a lot of geothermal energy.



RENEWABLE: HYDROELECTRIC

- **Hydroelectric power** is a way of harnessing energy from running water.
- Hydroelectric **dams** trap water in **reservoirs**
  - When electricity is needed, water is released and flows downwards with **gravity** to spin a **turbine**.
  - More reliable that solar and wind power.
  - Hydroelectric dams are very **expensive** and can



RENEWABLE: BIOFUEL

- **Biofuels** are made from **crops.**
- Ideally biofuels should be **carbon neutral** – absorb carbon dioxide as they grow and then give off carbon dioxide when they are burnt.
- However **fossil fuels are used in the production of biofuels**, for example in making fertilizers so they are not carbon neutral.
- Crops for biofuels could be used to feed people instead



HOW MUCH ENERGY DO I USE?

Power (watts) =
Energy (joules) ÷ Time (seconds)





4w
(charging)







100w







RECYCLING & ENERGY

- Everyday products like food containers, smartphones, computers and cars require a **lot of energy to produce**
- **Recycling** products **saves energy**
- The amount of energy you save depends on the product but it is **always worth recycling!**



Energy saved = 94%



Recycled aluminium can



Energy saved = 10-15%



Recycled glass jar

WHAT DO I REMEMBER?

1. _____ are natural substances required by humans for different needs.

2. Heat energy from the Earth is known as _____ energy.

3. _____ resources can be replaced and will not run out

4. Oil, coal, wind, sunlight and waves can be used to produce energy they are examples of _____.

Renewable

Energy resources

Coal

Non-renewable

Solar

Oil

Geothermal

Natural resources

WHAT DO I REMEMBER?

5. _____ is a non-renewable fuel formed from ancient land plants

6. _____ resources cannot be easily replaced, because they have taken millions of years to form.

7. _____ and gas are formed from marine microscopic plankton

8. The energy from sunlight is known as _____ energy and can be used to make electricity or heat.

Renewable

Energy resources

Coal

Non-renewable

Solar

Oil

Geothermal

Natural resources

Thank you!